SERANGOON JUNIOR COLLEGE
General Certificate of Education Advanced Level
Higher 2

CANDIDATE NAME	
CLASS	

CHEMISTRY — 9729/04

JC2 Preliminary Examination — **24 August 2018**

Paper 4 Practical — **2 hr 30 min**

Candidates answer on the Question Paper

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Give details of the practical shift and laboratory in the boxes provided.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough work.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.
Qualitative Analysis Notes are printed on pages 18 & 19.

At the end of the examination, fasten all your work securely together.
The number of marks is given in the brackets [ ] at the end of each question or part question.

Shift
Laboratory

For Examiner's Use	
1	/13
2	/29
3	/13
TOTAL	**/ 55**

This document consists of **19** printed pages and **1** blank pages.

Answer **all** the questions in the spaces provided.

1 Determination of a value for the solubility product, K_{sp}, of calcium iodate(V), $Ca(IO_3)_2$

Calcium iodate is used in the manufacture of disinfectants, antiseptics, and deodorants. Its solubility in water is low. When calcium iodate(V) is mixed with water, the following equilibrium is established.

$$Ca(IO_3)_2\ (s) \rightleftharpoons Ca^{2+}\ (aq) + 2IO_3^-(aq)$$

You are to perform an experiment to determine the solubility product, K_{sp}, of calcium iodate(V).

You will first prepare a saturated solution of calcium iodate(V) by mixing specific volumes of potassium iodate(V), KIO_3 and calcium nitrate, $Ca(NO_3)_2$.

The mixture is then filtered after leaving to stand for some time. The filtrate can then be analysed to determine the amount of iodate(V) ions left as follows:

- Excess potassium iodide, KI, is added to an acidified solution of the filtrate, liberating iodine.

$$IO_3^-(aq) + 5I^-(aq) + 6H^+\ (aq) \rightarrow 3I_2\ (aq) + 3H_2O\ (l) \quad \textbf{reaction 1}$$

- The liberated iodine is then titrated with a standard solution of sodium thiosulfate.

$$2S_2O_3^{2-}\ (aq) + I_2\ (aq) \rightarrow 2I^-\ (aq) + S_4O_6^{2-}\ (aq) \quad \textbf{reaction 2}$$

FA 1 0.200 mol dm^{-3} potassium iodate(V), KIO_3
FA 2 1.00 mol dm^{-3} calcium nitrate, $Ca(NO_3)_2$
FA 3 aqueous sodium thiosulfate, $Na_2S_2O_3$
FA 4 aqueous solution of potassium iodide, KI
FA 5 dilute hydrochloric acid, HC*l*
Starch indicator

(a) Preparing the reaction mixture

1. Transfer 50 cm^3 of **FA 1** using a measuring cylinder to the beaker labelled **reaction mixture**.
2. Using another measuring cylinder, transfer 20 cm^3 of **FA 2** into the same beaker.
3. A precipitate will form. Stir the mixture thoroughly and leave it to stand for several minutes to allow equilibrium to be reached.

While waiting, follow the instructions given in part (b).

(b) The concentration of **FA 1** provided is too high. You will first prepare a diluted solution of **FA 1** of known concentration and use it to standard the sodium thiosulfate solution provided.

1. Using a burette, transfer 10.00 cm^3 of **FA 1** into a 250 cm^3 graduated flask, labelled **diluted FA 1**.
2. Make up to the mark with deionised water and mix thoroughly.

Standardisation of FA 3

1. Fill a burette with **FA 3**.
2. Using a pipette, transfer 25.0 cm^3 of **diluted FA 1** into a conical flask.
3. Using a measuring cylinder, add about 10 cm^3 of **FA 4** to the flask.
4. Using another measuring cylinder, add about 2 cm^3 of **FA 5** to the flask.
5. Add **FA 3** from the burette into the flask until a pale yellow colour is obtained.
6. Add about 5 drops of starch solution into the flask. Continue adding **FA 3** until the blue-black colour just disappears.
7. Perform sufficient titrations to obtain accurate results for the end-point. Rinse the conical flask between each titration.

Record your titration results in the space below. Make certain that your recorded results show the precision of your working.

[2]

(iii) From your titration results, obtain a suitable volume of **FA 3** to be used in your calculations. Show clearly how you obtained this volume.

[1]

(iv) Calculate the concentration of $S_2O_3^{2-}$ ions in the **FA 3** solution.

[1]

Analysing the filtrate

(c) **(i)** 1. Filter the reaction mixture through a dry filter paper into a dry conical flask, labelled **FA 6**. This is the filtrate, **FA 6**. Do not wash the white precipitate with water.

2. Pipette 10.0 cm^3 of **FA 6** into a conical flask.
3. Using a measuring cylinder, add about 10 cm^3 of FA 4 to the flask.
4. Using another measuring cylinder, add about 2 cm^3 of **FA 5** to the flask.
5. Add **FA 3** from the burette into the flask until a pale yellow solution is obtained.
6. Add about 5 drops of starch indicator and continue adding **FA 3** until the blue-black colour just disappears.
7. Perform sufficient titrations to obtain accurate results for the end-point. Rinse the conical flask between each titration.

Record your titration results in the space below. Make certain that your recorded results show the precision of your working.

(ii) From your titration results, obtain a suitable volume of **FA 3** to be used in your calculations. Show clearly how you obtained this volume.

Calculations

(d) **(i)** Calculate the amount of $S_2O_3^{2-}$ ions present in the volume of **FA 3** obtained in **(c)(ii)**.

[1]

(ii) Calculate the amount of IO_3^- ions present in 10.0 cm^3 of the filtrate, **FA 6**.

[1]

(iii) Hence, calculate the total amount of IO_3^- ions present in the filtrate, **FA 6**.

[1]

(e) **(i)** Calculate the initial amount of IO_3^- ions and Ca^{2+} ions present in the reaction mixture prepared in **(a)**.

[1]

(ii) Calculate the amount of IO_3^- ions precipitated as $Ca(IO_3)_2$.

[1]

(iii) Hence, calculate the amount of Ca^{2+} ions left in **FA 6**.

[1]

(f) **(i)** Use your answer in parts **(d)(iii)** and **(e)(iii)** to calculate a value for the solubility product, K_{sp}, of calcium iodate(V). Include units in your answer.

[1]

(g) Another student performed this experiment and obtained a value for the solubility product, K_{sp}, of 3.45×10^{-5}. A literature value for this solubility product is 6.71×10^{-6} at 20 °C.

You should assume that apparatus of the same precision was used in each case.

State a possible reason for the higher value of K_{sp} obtained by the student and suggest an improvement which might allow a value closer to the literature value to be obtained.

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[2]

[Total: 13]

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2 You are to determine the enthalpy change of reaction, ΔH, for the reaction shown below.

$$Cu(s) + H_2SO_4(aq) \rightarrow CuSO_4(aq) + H_2(g)$$

As copper is an unreactive metal it does not react with dilute acids. You will need to find the enthalpy change of reaction for two reactions that do occur. The equations for these two reactions are below.

$$Mg(s) + H_2SO_4(aq) \rightarrow MgSO_4(aq) + H_2(g) \qquad \textbf{Reaction 1}$$

$$Mg(s) + CuSO_4(aq) \rightarrow MgSO_4(aq) + Cu(s) \qquad \textbf{Reaction 2}$$

You will conduct experiments to find the enthalpy changes for each of **Reaction 1** and **Reaction 2** and use these values to calculate the enthalpy change for the reaction of copper with sulfuric acid.

Determining the enthalpy change for Reaction 1

$$Mg(s) + H_2SO_4(aq) \rightarrow MgSO_4(aq) + H_2(g) \qquad \textbf{Reaction 1}$$

(a) Method

FA 8 is 1.00 mol dm^{-3} sulfuric acid, H_2SO_4.
FA 7 is magnesium powder, Mg.

Read through the method before you start any practical work and prepare a suitable table for your results.

- Weigh the bottle containing **FA 7**. Record the mass.
- Support the Styrofoam cup in the 250 cm^3 beaker.
- Use the measuring cylinder to transfer 25 cm^3 of **FA 8** into the Styroofoam cup.
- Measure the temperature of **FA 8** in the cup and start the stop clock. Record this temperature as being the temperature at time = 0.
- Measure, and record, the temperature of this **FA 8** every half minute for 2 minutes.
- At time = 2½ minutes add the **FA 7** to the acid and stir carefully to reduce acid spray.
- Measure the temperature of the mixture in the cup at time = 3 minutes and then every half minute up to time = 7 minutes.
- Continue stirring occasionally throughout this time.
- Weigh the bottle that had contained **FA 7**. Record the mass.
- Calculate and record the mass of **FA 7** added to the sulfuric acid.
- Rinse the cup with water and shake to dry.

Recording

[6]

(b) (i) On the grid below plot a graph of temperature (*y*-axis) against time (*x*-axis).

[2]

(ii) Complete the graph by drawing two, straight lines of best fit

- One to show the temperature up to time = 2 ½ minutes
- One to show the temperature after time = 2 ½ minutes

[1]

(iii) From your graph, use the two straight lines of best fi t to calculate the change in temperature at time = 2 ½ minutes.

Temperature change = ………………^{o}C

[1]

(c) Calculations

(i) In the reaction in **(a)**, the sulfuric acid was in excess. Without carrying out any additional tests, what observation could you have made during your experiment to confirm this?

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[1]

(ii) Calculate the energy change that occurred during the reaction in **(a)**.
[Assume that 4.2 J is needed to raise the temperature of 1.0 cm^3 of solution by 1.0 ^oC.]

Energy change = …………………J
[1]

(iii) Use your answer to **(ii)** to calculate the enthalpy change, in kJ mol^{-1}, for the reaction between sulfuric acid and magnesium.
[*A*r : Mg, 24.3]

$Mg(s) + H_2SO_4(aq) \rightarrow MgSO_4(aq) + H_2(g)$ **Reaction 1**

Enthalpy change of Reaction 1 = ………………..kJ mol^{-1}
[2]

(d) Planning: Determining the enthalpy change for Reaction 2

The enthalpy change of reaction for **Reaction 2** can be determine using a **non-graphical approach**.

$$Mg(s) + CuSO_4(aq) \rightarrow MgSO_4(aq) + Cu(s) \qquad \textbf{Reaction 2}$$

(i) Plan an investigation to determine mean temperature rise, mean mass of magnesium used and the enthalpy change of reaction for **Reaction 2**.

You are provided with

- **FA 11**, 1.00 mol dm^{-3} copper(II) sulfate, $CuSO_4$
- **FA 9** and **FA 10** magnesium powder samples in weighing bottles
- the equipment normally found in a school or college laboratory.

You are to take precaution to ensure that the copper(II) sulfate was in excess in these reactions and **not more than 30 cm^3** is used for each experiment.

In your plan you should include brief details of

- the volume of $CuSO_4$ you plan to use (you may assume the mass of Mg(s) used to be the same as in reaction 1),
- the apparatus that you would use,
- the procedure that you would follow and the measurements that you would take,
- a suitable table for your results.

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[6]

(ii) **Perform** the experiment you have planned in **(d)(i)**, clearly calculating the

- mean temperature change and
- mean mass of magnesium used.

Hence, determine the enthalpy change, in kJ mol^{-1} for the reaction between magnesium and copper(II) sulfate.

[Assume that 4.2 J is needed to raise the temperature of 1.0 cm^3 of solution by 1.0^oC]
[A_r : Mg, 24.3]

[4]

Enthalpy change for Reaction 3

Reaction 3 is shown below.

$Cu(s) + H_2SO_4(aq) \rightarrow CuSO_4(aq) + H_2(g)$ **Reaction 3**

(e) Use your values for the enthalpy changes for **Reactions 1** and **2** to calculate the enthalpy change for **Reaction 3**.

$Mg(s) + H_2SO_4(aq) \rightarrow MgSO_4(aq) + H_2(g)$ **Reaction 1**

$Mg(s) + CuSO_4(aq) \rightarrow MgSO_4(aq) + Cu(s)$ **Reaction 2**

Show clearly how you obtain your answer.

(If you were unable to calculate the enthalpy changes for **Reactions 1** and **2**, you should assume that the value for **Reaction 1** is – 444 kJ mol^{-1} and that the value for **Reaction 2** is – 504 kJ mol^{-1}. Note: these are not the correct values.)

Enthalpy change for **Reaction 3** = …………………kJ mol^{-1}

[2]

(f) **(i)** The method you used to determine the enthalpy change for **Reaction 1** was more accurate than the method you used to determine the enthalpy change for **Reaction 2**. Suggest **two** reasons why the method used for **Reaction 2** was less accurate. Explain your answers.

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[2]

(ii) A student suggested that the accuracy of the method used for **Reaction 2** could be improved by using a larger volume of copper(II) sulfate. Is this a correct suggestion? Give a reason for your answer.

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[1]

[Total: 29]

3 **FA 12** contains two cations and two anions from those listed on pages 18 and 19.

In all tests, the reagent should be added gradually until no further change is observed, with shaking after each addition.
Record your observations in the spaces provided.
Your answers should include
• details of colour changes and precipitates formed,
• the names of gases evolved and details of the test used to identify each one.

You should indicate clearly at what stage in a test a change occurs, writing any deductions you made alongside the observations on which they are based.

Marks are **not** given for chemical equations.

No additional or confirmatory tests for ions present should be attempted.

Candidates are reminded that definite deductions may be made from tests where there appears to be no reaction.

	Test	*Observations* [3]	*Possible Cation* [3]
(a)	Place 3 cm^3 **FA 12** in a test tube, add aqueous sodium hydroxide, drop by drop, until no further change is seen.		
(b)	Filter the mixture from **(a)** and **collect the filtrate for later tests** Leave the residue in the filter paper and observe it again after several minutes.		
(c)	Extract about 1 cm^3 of the filtrate using a teat pipette into a clean test tube. Add dilute nitric acid, drop by drop, until no further change is seen.		

Test for anions

Test **(e)** has been conducted and the observation is recorded.

You are to complete test **(f)** and devise **two** more tests to confirm the identity of the anions present in **FA 12**.

In the space provided below, clearly state the tests, observations and deductions.

	Test [2]	*Observation* [3]
(e)	Add 1 cm^3 of **FA 12** into a clean test-tube. Add 2 cm^3 of aqueous hydrochloric acid.	No effervescence observed
(f)	Add 1 cm^3 of **FA 12** into a clean test-tube Add 3 drops of aqueous silver nitrate.	
(g)		
(h)		

Anions present in **FA12**: ………………and …………………. [1]

[Total: 13]

Qualitative Analysis Notes
[ppt. = precipitate]
(a) Reactions of aqueous cations

cation	***reaction with***	
	NaOH(aq)	NH_3(aq)
aluminium, Al^{3+}(aq)	white ppt. soluble in excess	white ppt. insoluble in excess
ammonium, NH_4^+(aq)	ammonia produced on heating	–
barium, Ba^{2+}(aq)	no ppt. (if reagents are pure)	no ppt.
calcium, Ca^{2+}(aq)	white ppt. with high [Ca^{2+}(aq)]	no ppt.
chromium(III), Cr^{3+}(aq)	grey-green ppt. soluble in excess giving dark green solution	grey-green ppt. insoluble in excess
copper(II), Cu^{2+}(aq),	pale blue ppt. insoluble in excess	blue ppt. soluble in excess giving dark blue solution
iron(II), Fe^{2+}(aq)	green ppt., turning brown on contact with air insoluble in excess	green ppt., turning brown on contact with air insoluble in excess
iron(III), Fe^{3+}(aq)	red-brown ppt. insoluble in excess	red-brown ppt. insoluble in excess
magnesium, Mg^{2+}(aq)	white ppt. insoluble in excess	white ppt. insoluble in excess
manganese(II), Mn^{2+}(aq)	off-white ppt., rapidly turning brown on contact with air insoluble in excess	off-white ppt., rapidly turning brown on contact with air insoluble in excess
zinc, Zn^{2+}(aq)	white ppt. soluble in excess	white ppt. soluble in excess

(b) Reactions of anions

anion	*reaction*
carbonate, CO_3^{2-}	CO_2 liberated by dilute acids
chloride, Cl^-(aq)	gives white ppt. with Ag^+(aq) (soluble in NH_3(aq))
bromide, Br^-(aq)	gives pale cream ppt. with Ag^+(aq) (partially soluble in NH_3(aq))
iodide, I^-(aq)	gives yellow ppt. with Ag^+(aq) (insoluble in NH_3(aq))
nitrate, NO_3^-(aq)	NH_3 liberated on heating with OH^-(aq) and A*l* foil
nitrite, NO_2^-(aq)	NH_3 liberated on heating with OH^-(aq) and A*l* foil; NO liberated by dilute acids (colourless NO → (pale) brown NO_2 in air)
sulfate, SO_4^{2-}(aq)	gives white ppt. with Ba^{2+}(aq) (insoluble in excess dilute strong acids)
sulfite, SO_3^{2-}(aq)	SO_2 liberated with dilute acids; gives white ppt. with Ba^{2+}(aq) (soluble in dilute strong acids)

(c) Tests for gases

gas	*test and test result*
ammonia, NH_3	turns damp red litmus paper blue
carbon dioxide, CO_2	gives a white ppt. with limewater (ppt. dissolves with excess CO_2)
chlorine, Cl_2	bleaches damp litmus paper
hydrogen, H_2	"pops" with a lighted splint
oxygen, O_2	relights a glowing splint
sulfur dioxide, SO_2	turns aqueous acidified potassium manganate(VII) from purple to colourless

(d) Colour of halogens

halogen	*colour of element*	*colour in aqueous solution*	*colour in hexane*
chlorine, Cl_2	greenish yellow gas	pale yellow	pale yellow
bromine, Br_2	reddish brown gas / liquid	orange	orange-red
iodine, I_2	black solid / purple gas	brown	purple

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